

Fundamentals of Information Systems Security

Chapter 5 Quiz Answer Key

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Section A: Multiple Choice (40 points)

1. **B** — Identification, Authentication, Authorization, Accountability
2. **B** — Discretionary Access Control (DAC)
3. **A** — The point where false rejection rate equals false acceptance rate
4. **C** — Smart card
5. **B** — Giving users the minimum access necessary to perform their job
6. **B** — Mandatory Access Control (MAC)
7. **A** — Multi-Factor Authentication
8. **C** — RADIUS
9. **A** — The central part of a computing environment that enforces access control
10. **B** — Identification is claiming identity; authentication is proving identity

Section B: True/False (20 points)

1. **False** — Physical access controls control physical resources like buildings
2. **True** — Biometrics can be used for both
3. **True** — SSO allows single login to multiple systems
4. **True** — Kerberos provides mutual authentication
5. **False** — Bell-LaPadula focuses on confidentiality
6. **True** — ACLs define permissions on objects
7. **False** — Biba focuses on integrity, not confidentiality
8. **False** — Cloud computing still requires access controls
9. **True** — Accountability traces actions to individuals
10. **True** — Temporal isolation restricts access by time

Section C: Short Answer (40 points)

1. DAC, MAC, and RBAC (10 points)

- **DAC (Discretionary Access Control):** Owner of resource decides access; permissions can be transferred. *Example: File permissions in Windows*
- **MAC (Mandatory Access Control):** System enforces access based on security labels and clearances; permissions cannot be transferred. *Example: Military classified systems*
- **RBAC (Role-Based Access Control):** Access is based on job roles; users are assigned to roles. *Example: Enterprise HR systems*

2. Three Authentication Factors (10 points)

- **Something you know:** Password, PIN, passphrase, security questions
- **Something you have:** Smart card, key, badge, token, phone
- **Something you are:** Fingerprint, iris scan, voice pattern, facial recognition

3. SSO Advantages and Disadvantages (10 points)

- **Advantages:** Efficiency (single login), stronger passwords (only one to remember), centralized administration, continuous session monitoring
- **Disadvantages:** Catastrophic compromise (one password gives access to all), single point of failure, integration challenges with legacy systems

4. Kerberos Authentication (10 points)

- User sends ID and access request through Kerberos client to KDC
- Authentication server at KDC verifies user and requested service in KDC database
- KDC sends a ticket (user's unique time-stamped key)
- Ticket is used to access the requested service
- **KDC (Key Distribution Center):** Central component that issues tickets and manages authentication; must be physically secured and hardened